Technical Study Notes: SQL Functions and Operators (Chapters 3 & 4)

1. Overview of SQL Functions (Chapter 3)

Oracle SQL provides an extensive library of built-in functions to transform and manipulate data.

- Total Functions: 300+ available functions.
- Standard Usage: Approximately 45 functions are used in common database operations.
- Functional Hierarchy:
    - Level 1: Single Row (Scalar) Functions: Operate on single rows and return one result per row.
    - Level 2: Multi-row (Group) Functions: Operate on sets of rows to return a single aggregate result.

Types of Single Row Functions

Single row functions are categorized into five functional groups:

1. Character: For text string manipulation.
2. Number: For mathematical operations.
3. Date: For time-based calculations and formatting.
4. Conversion: For changing data between types (e.g., String to Date).
5. General: Logic-based functions (Null handling and conditionals).

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2. Character Functions (Chapter 3)

Manipulation refers to performing any type of change to character data.

2.1 Case Manipulation Functions

These functions alter the capitalization of alphabetic characters.

- UPPER: Converts all alphabets to capital letters.
- LOWER: Converts all alphabets to small/lowercase letters.
- INITCAP: Converts the first alphabet of each word to capital and the rest to small.
    - Word Separator Rule: INITCAP uses special characters (e.g., spaces, underscores, hyphens) as delimiters to identify new words.

Technical Note: Oracle recognizes three character types: Alphabets (A-Z, a-z), Digits (0-9), and Special Characters (all others). There is no specific function for "Mixed Case"; mixed formatting must be achieved through nested manipulation.

2.2 Character Manipulation Functions

- CONCAT: Joins exactly two arguments. Functionally identical to the || pipe operator.
- LPAD / RPAD: Adds characters to the left (LPAD) or right (RPAD) to reach a specified length. The default 3rd argument is a space.
- LTRIM / RTRIM: Removes specific characters from the left or right of a string.
- TRIM: Removes a single character from the left, right, or both sides (defaults to both).
- LENGTH: Returns the numeric count of characters in a string.
- SUBSTR: Extracts a specific portion of a string (from left, middle, or right).
- INSTR: Finds the numeric position of a character or substring within a string.
- TRANSLATE vs. REPLACE:
    - TRANSLATE: Operates character-by-character. Requires exactly three arguments. If you translate 'ABC' to '123', every 'A' becomes '1', every 'B' becomes '2', and every 'C' becomes '3'.
    - REPLACE: Operates word-by-word (string-by-string). It looks for the exact sequence of the search string to replace it with the replacement string.

2.3 Nested Functions

Nested functions are "functions within functions."

- Execution Rule: The innermost function is evaluated first. Its result is then passed as an input argument to the next outer function.

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3. Number Functions and Numeric Logic (Chapter 3)

List of Core Number Functions

- POWER: Calculates "raised to" or "root" values. If the 2nd argument is ≥1, it returns the raised value; if <1, it returns the root.
- SQRT: Returns the square root of a positive number.
- MOD: Returns the remainder of a division operation.
- FLOOR: Returns the same integer or the previous lower integer.
- CEIL: Returns the same integer or the next higher integer.
- TRUNC: Removes digits after the decimal point.
    - Technical Detail: If the second argument is negative, TRUNC converts digits before the decimal point to zero (e.g., TRUNC(1234, -2) results in 1200).
- ROUND: Follows Base-10 rounding logic.

Rounding Logic (Base 10)

- First Half (0-4): No change is made to the preceding digit.
- Second Half (5-9): Add 1 to the preceding digit.

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4. SQL Operators and Pattern Matching (Chapter 3)

Operators facilitate filtering and comparison within the WHERE clause.

- Single Row Operators: =, <, >, <=, >=, <>, !=, ^=
- Multi-row Operators: ANY, SOME, ALL
- Special Operators: IN, BETWEEN, LIKE, IS NULL
- Logical Operators: NOT, AND, OR

Pattern Matching Wildcards (Used with LIKE)

- _ (Underscore): Represents exactly one single character.
- % (Percent): Represents zero or more characters.

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5. Date Functions and Date Arithmetic (Chapter 3)

The default Oracle date format is DD-MON-RR.

5.1 Core Date Functions

- SYSDATE: Returns the current date and time from the Database Server.
- CURRENT_DATE: Returns the date and time from the Local/Client machine.
- ADD_MONTHS: Adds/subtracts months. Used for larger intervals:
    - 1 Decade = 120 Months
    - 1 Century = 1200 Months
- MONTHS_BETWEEN: Returns the numeric count of months between two dates.
- LAST_DAY: Returns the last date of the input month.
- NEXT_DAY: Returns the date of the next specific weekday (e.g., 'FRIDAY').

5.2 Date Arithmetic

A date contains 7 parts: Century, Year, Month, Day, Hour, Minute, and Second.

- Valid Operations: Date + Interval, Date - Interval, Date - Date.
- Constants: 1 Day = 24 Hours = 1440 Minutes = 86400 Seconds.
- Intervals: 1 Week = 7 Days; 1 Quarter = 3 Months.

5.3 Date TRUNC and ROUND Logic

Operations are typically performed against the intervals MONTH, Q (Quarter), and YEAR.

TRUNC Logic: Always returns the first date of the current interval (e.g., TRUNC(date, 'YEAR') is always Jan 1st).

ROUND Logic (First Half vs. Second Half Matrix): | Interval | First Half (Returns Start of Current) | Second Half (Returns Start of Next) | | :--- | :--- | :--- | | MONTH | Day 1 to 15 | Day 16 to 31 | | QUARTER (Q1) | Jan 1 to Feb 15 | Feb 16 to Mar 31 | | QUARTER (Q2) | Apr 1 to May 15 | May 16 to Jun 30 | | QUARTER (Q3) | Jul 1 to Aug 15 | Aug 16 to Sep 30 | | QUARTER (Q4) | Oct 1 to Nov 15 | Nov 16 to Dec 31 | | YEAR | Jan 1 to Jun 30 | Jul 1 to Dec 31 |

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6. General and Conditional Functions (Chapter 4)

6.1 Null-Related Functions

- NVL(arg1, arg2): If arg1 is null, returns arg2.
- NVL2(arg1, arg2, arg3): If arg1 is NOT null, returns arg2. If arg1 IS null, returns arg3.
    - Critical Data Type Rule: If arg2 is character-based, arg3 can be any type. If arg2 is non-character, arg3 must match the data type of arg2.
- NULLIF(arg1, arg2): Returns NULL if both arguments are equal; otherwise, returns arg1.
- COALESCE(arg1...argN): Returns the first non-null value in a list of infinite arguments. All arguments must share the same data type.

6.2 Conditional Logic

- DECODE / CASE I: Best for simple equality checks.
- CASE II: Supports complex conditions and range-based evaluations (e.g., SALARY BETWEEN 0 AND 6000), which is an advantage over DECODE.

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7. Conversion Functions and Format Models (Chapter 4)

7.1 Conversion Directions

- TO_NUMBER: Character → Number (for calculations/comparisons).
- TO_DATE: Character → Date (for arithmetic/comparisons).
- TO_CHAR: Number/Date → Character (for custom display formatting).

7.2 Number Format Elements

- 9 (Digit), 0 (Leading/Trailing Zero), . (Decimal), , (Group separator).
- NLS Parameters:
    - D: Decimal point (Default .).
    - G: Group separator (Default ,).
    - L: Local Currency (NLS_CURRENCY).
    - U: Dual/Euro Currency (NLS_DUAL_CURRENCY). Pro-tip: Use Alt+157 for the € symbol.
- S: Sign (+/-), MI: Negative sign after number, PR: Negative numbers in <brackets>.
- EEEE: Scientific notation (e.g., 2.40E+04 which represents 2.4×10
- 4
-).
- V: Multiplication factor. It multiplies the value by 10
- n
- , where n is the number of 9s appearing after the V.

7.3 Date Format Elements

- DD (Day), MM (Month #), MON/MONTH (Name), RM (Roman Month).
- Century Logic:
    - YY: Uses the current century (e.g., 95 becomes 2095).
    - RR: 50-year split logic. 00-49 → Current Century; 50-99 → Previous Century.
- J (Julian Day): Number of days since Jan 1, 4712 BC.

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8. Substitution Variables and Session Settings (Chapter 4)

- & (Single Ampersand): Prompts for a value, uses it once, and discards it.
- && (Double Ampersand): Prompts once and preserves the value for the entire session.
- SET VERI OFF: Session command used to suppress the "old/new" substitution messages.

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9. Practical Application Scenarios

Character and Number Scenarios

- Name Formatting: Displaying mixed inputs like stEvEn as STeVeN using nested UPPER, LOWER, and SUBSTR.
- Consonant Count: Using TRANSLATE or REPLACE to remove vowels and LENGTH to count the remaining consonants.
- Validation: Finding employees with 5-figure salaries using LENGTH(SALARY) = 5.

Date Calculation Scenarios

- Tenure Tracking: Finding the number of days, weeks, or quarters worked using date arithmetic.
- Joining Period: Calculating the number of days worked specifically within the employee's joining month using LAST_DAY.
- Julian Utilities: Using TO_CHAR(SYSDATE, 'SSSSS JSP') to display the current second of the day in words.

Business Logic Scenarios

- Payroll Scheduling:
    - Finding the "4th Saturday after joining" for a bonus.
    - Finding the "3rd Wednesday of the next month" for standard pay dates.
    - Finding the "Friday after the 10th of the month" for specialized disbursements.
- Conditional Increments: Using CASE II to apply a 40% raise for salaries between 0-6000 and 30% for 6001-9000.
- Managerial Hierarchy: Using NVL(MANAGER_ID, 'BOSS') or NVL2(MANAGER_ID, 'Employee', 'Boss') to handle organizational reporting.